

Bitter melon extract impairs prostate cancer cell cycle progression and delays prostatic intraepithelial neoplasia in TRAMP model

Prostate cancer remains the second leading cause of cancer deaths among American men. Earlier diagnosis increases survival rate in patients. However, treatments for advanced disease are limited to hormone ablation techniques and palliative care. Thus, new methods of treatment and prevention are necessary for inhibiting disease progression to a hormone refractory state. One of the approaches to control prostate cancer is prevention through diet, which inhibits one or more neoplastic events and reduces the cancer risk. For centuries, Ayurveda has recommended the use of bitter melon (*Momordica charantia*) as a functional food to prevent and treat human health related issues. In this study, we have initially used human prostate cancer cells, PC3 and LNCaP, as an in vitro model to assess the efficacy of bitter melon extract (BME) as an anti-cancer agent. We observed that prostate cancer cells treated with BME accumulate during the S phase of the cell cycle, and modulate cyclin D1, cyclin E and p21 expression. Treatment of prostate cancer cells with BME enhanced Bax expression, and induced poly(ADP-ribose) polymerase cleavage. Oral gavage of BME, as a dietary compound, delayed the progression to high grade prostatic intraepithelial neoplasia (PIN) in TRAMP (transgenic adenocarcinoma of mouse prostate) mice (31%). Prostate tissue from BME-fed mice displayed ~51% reduction of PCNA expression. Together, our results suggest for the first time that oral administration of BME inhibits prostate cancer progression in TRAMP mice by interfering cell cycle progression and proliferation.